

name: k. Ramesh

course code: DSA0302

reg no: 192472385

CO2_AT2 - scenario based

1) morphological processing strategy

Morphological processing is the process of analyzing internal structure of words by identifying root words, prefixes and suffixes

objective:

* decomposes each word into root and affixes

* normalizes all variants to a common base form for efficient indexing and retrieval

Read the input words:

analyzing

analysis

analytical

Detect morphological patterns

check for prefixes

check for suffixes

identify root word

Classify the transformation

if only the grammatical form changes - inflectional

if meaning or word class changes - derivational

normalize all words to common representation

Apply morphological rules

analyzing:

word end with -ing

remove -ing

Base word analyze

analysis

word end with -sis

related base word analyze

analytical

word end with -ical

related base word analyze

Tagging decision

word
analyzing

morphological type

reason

inflectional

only grammatic form changes

analysis

Derivational

verb changes to noun

analytical

Derivational

verb changes to adjective

Normalization

original	normalized form
analyzing	analyze
analysis	analyze
analytical	analyze

structured report

original	root	affix	type	normalized
analyzing	analyze	-ing	inflectional	analyze
analysis	analyze	-sis	derivational	analyze
analytical	analyze	-ical	derivational	analyze

justification

-ing changes only grammatical form

-sis creates noun

-ical creates an adjective

Output:

the search engine stores all three words under common representation

2) morphological parsing strategy

read input words

detect prefixes and suffixes using rule based analysis

identify root word

normalize all words base form agree

Apply process

disagree

original: disagree

prefix: dis

root word: agree

suffix: none

transformation: derivational

the prefix changes meaning so it

derivational transformation

agreeable

original: agreeable

prefix: none

root: agree

suffix: able

transformation: derivation

the suffix changes word class from verb to adjective

structured output

original	prefix	root	suffix	transformation	semantic interpretation	normalized
disagree	dis	agree	-	Derivational	opposite	agree
agreement	-	agree	-ment	Derivational	noun	agree
agreeable	-	agree	-able	Derivational	adjective	agree

agreement

original: agreement

prefix: none

root: agree

suffix: -ment

transformation: derivation

* the suffix changes word

class from verb to noun

All words are normalized to agree allowing the sentiment analysis system identify related words while preserving their meaning

3) morphology Based normalization mechanism

read input words

Detect derivational suffixes

identify derivational level

normalize all words to common root govern

Apply process

govern

root: govern

affix: none

derivational: root

normalized: govern

government

root: govern

affix: -ment

derivational: level-1

normalized: govern

governance

root: govern

affix: -ance

derivational: level 1

normalized: ~~govern~~

Output:

original	root	affix	Derivational	normalized
govern	govern	-	root	govern
government	govern	-ment	level 1	govern
governance	govern	-ance	level 1	govern

result:

All words are normalized to govern improving topic modeling and document clustering

4) morphological parsing strategy

read input words

detect prefix and suffix

analyze effect on meaning and word class

normalize base representation act

Apply process

activate

prefix: none

root: act

suffix: iate

Derivational: act-activate

word: verb

normalized: act

activation

prefix: none

root: act

suffix: -ion

Derivational: act-activate-activation

word: noun

meaning: process making active

normalize: act

reactivation

prefix: re-

root: act

suffix: -ion

Derivational: act-activate-activation
-re-activation

word: noun

meaning: Activate again

normalized: act

Output

original	prefix	root	suffix	Derivational	normalized
activate	-	act	ivate	act-activate	act
activation	-	act	ion	act-activate-activation	act
reactivation	re	act	ion	act-activate-activation -re-activation	act

result:

The system maps all words the common base act improving document classification and semantic indexing

5) inflectional morphology-based normalization

read input words

detect grammatic suffixes

classify each grammatical form

normalize all word to create

Apply process

create

root: create

suffix: none

grammatical: Base form

normalized: create

creates

root: create

suffix: -s

grammatical: third person

singular present

normalized: create

creating

root: create

suffix: -ing

grammatical: present participle

normalized: create

output:

original	suffix	grammatical	root	normalized
create	-	Base form	create	create
creates	-s	third person singular	create	create
creating	-ing	present participle	create	create

result:

All grammatical forms are normalized to create, improving search optimization and information retrieval while preserving original meaning

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